Dyslexia at age 11 years.

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In 1957, when cohort members were aged 11 years, 4281 cohort members were administered tests of general cognitive ability, arithmetic, reading and vocabulary. These tests were developed by National Foundation for Educational Research in England and Wales (Pigeon, 1964). Dyslexia status in was based on the results of the general ability and reading ability tests. All tests were administered by teachers in a school setting.

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Document details

Categories of variables	Cognitive Function
Summary of work undertaken	Derivation of a variable measuring dyslexia status at age 11. Calculated using GA11N [Normalised score of 80-item verbal/non-verbal ability test at 11yrs] and R11r [Reading ability at 11 yrs].
Output variables	dyslex11

Summary

In 1957, when cohort members were aged 11 years, 4281 cohort members were administered tests of general cognitive ability, arithmetic, reading and vocabulary. These tests were developed by National Foundation for Educational Research in England and Wales (Pigeon, 1964). Dyslexia status in was based on the results of the general ability and reading ability tests. All tests were administered by teachers in a school setting.

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Raw variables

GA11N – Normalized score of the general ability test. The general ability test consists of 56 exercises (half verbal and half non-verbal), each representing a logical order of words or figures; in each case the task is to select one of five items to complete the sequence. The total score was normalized to 100 and an SD of 15 to give a score comparable to IQ. R11R – Raw score of the reading ability test. Test consists of 50 different words to read and pronounce, ranging from low difficulty (e.g., egg, book, apple) to high difficulty (e.g., extraneous, ophthalmic). These variables can be picked up directly from $lb/d03/c.dat

Derived variables

DYSLEX11 – Dyslexia status at age 11 years. Categorical variable. Children were classified as dyslexic if they achieved a GA11N score of 70 or higher, and a R11R score lower than 20.66, which is 1.5 SD below the mean (see the NB in the syntax file). Children who scored lower than these thresholds on both tests were classified as having low ability. The group scoring at or above threshold on both tests provided a normal comparison (reference) group. 999 coded as missing.

Variable info

DYSLEX11

Purpose : Dyslexia status **at 11 years** Year : 1957

1. Definition

Value	Label
1	Normal ability
2	Dyslexia
3	Low ability
999	Missing data

2. Specification

Dyslexia status at age 11 has been derived using scores from reading and verbal/non-verbal ability tests undertaken in 1957 (at 11 years)

The threshold for R11R was defined as 1.5 SD below mean. SD of R11R =10.51 and Mean=36.43. So threshold=36.43-(10.51*1.5) =20.66

Children were classified as dyslexic if they achieved a GAIIN score of 70 or higher and a R11R score lower than 20.66. The threshold of 70 for GA11N represents the cut-off for mild learning disability. Anything above this is considered within the normal range for general cognitive ability/IQ.

3. Derivation

COMPUTE DYSLEX11 = 999. IF (GA11N GE 70 AND R11R GE 20.66) DYSLEX11=1. IF (GA11N GE 70 AND R11R LT 20.66) DYSLEX11=2. IF (GA11N LT 70 AND R11R LT 20.66) DYSLEX11=3. IF (GA11N E SYSMIS or R11R E SYSMIS) DYSLEX11=999.

History of cleaning/derivation/documentation

27/03/2009 - Derivation of Dyslexia status at age 11 (1957)

Variables in this document

1 high-confidence match

Variable	Label	How it was found
Clinical data [51] › Cognitive function [101] › Childhood cognition [1011]		
dyslex11	Dyslexia status - at age 11 years	topsheet field

1 medium-confidence match

Variable	Label	How it was found
Clinical data [51] › Cognitive function [101] › Childhood cognition [1011]		
ga11n	Verbal and non-verbal ability - General Ability (80-items - normalised score) - at age 11 years	prose scan

1 low-confidence match

Variable	Label	How it was found
Clinical data [51] › Cognitive function [101] › Childhood cognition [1011]		
r11r	Mechanical reading - pronunciation - (edited raw score) - at age 11 years	prose scan